AuraDB Knowledge Graph Creation

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1. For each programme, how many distinct courses there are?

```
1 MATCH (p:StudyProgramme)-[:HAS_COURSE]->(c:Course)
2 RETURN p.title AS programme,
3        count(DISTINCT c) AS number_of_courses
4 ORDER BY number_of_courses DESC;
```

Run

Table Raw

Search table...

Graph chip

	programme	number_of_courses
1	"Digital Humanities"	50
2	"Business Informatics"	29

Started streaming 2 records after 22 ms and completed after 23 ms.

2. Which courses do both programmes share?

```
1 MATCH (p:StudyProgramme)-[:HAS_COURSE]->(c:Course)
2 WITH c, collect(DISTINCT p.title) AS programmes
3 WHERE size(programmes) = 2
4 RETURN c.course_id AS course_id,
5        c.title AS shared_course,
6        programmes
7 ORDER BY shared_course;
```

Run

Table Raw

Search table...

Graph chip

course_id	shared_course	programmes
1 "DE0729"	"Communication and Presentation Skills"	["Business Informatics", "Digital Humanities"]
2 "DE0739"	"Knowledge Management Systems"	["Business Informatics", "Digital Humanities"]
3 "DE0645"	"Portfolio Management Technologies"	["Business Informatics", "Digital Humanities"]
4 "DE0751"	"e-Business Solutions"	["Business Informatics", "Digital Humanities"]

System Settings

Started strea ms and completed after 31 ms.

3. Topics which are covered in Knowledge Management Systems?

```
1 MATCH (c:Course {course_id: 'DE0739'})-[:COVERS_TOPIC]->(t:Topic)
2 RETURN c.title AS course,
3        t.ordinal AS topic_number,
4        t.title AS topic
5 ORDER BY topic_number;
```

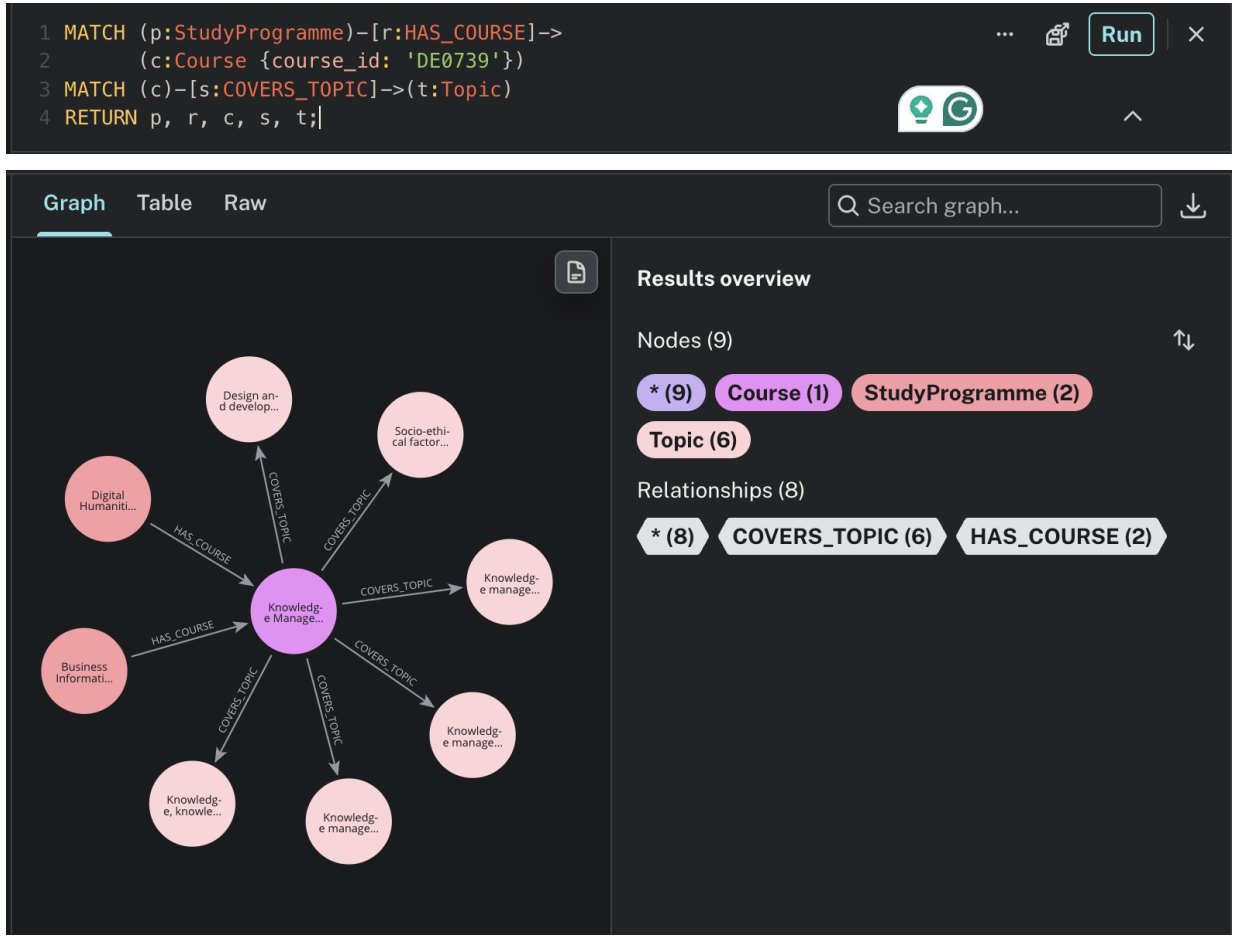
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	course	topic_number	topic
1	"Knowledge Management Systems"	1	"Knowledge, knowledge management aims, perspectives and challenges."
2	"Knowledge Management Systems"	2	"Knowledge management methods and systems."
3	"Knowledge Management Systems"	3	"Knowledge management processes and tools (including sub-topics on knowledge audit and user experience design)."
4	"Knowledge Management Systems"	4	"Knowledge management relation to strategic, human resources, innovation and quality management."
5	"Knowledge Management Systems"	5	"Socio-ethical factors of knowledge management."
6	"Knowledge Management Systems"	6	"Design and development of knowledge management system's blueprint."

Started streaming 6 records after 28 ms and completed after 29 ms.

4. Graph showcasing how both programmes are connected to this course and topics.



5. Learning outcomes of the course?

```
MATCH (c:Course {course_id: 'DE0739'})-[:HAS_OUTCOME]->(o:LearningOutcome)
RETURN o.ordinal AS outcome_number,
       o.text AS learning_outcome
ORDER BY outcome_number;
```

Run

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	outcome_number	learning_outcome
1	1	"Is able to perform a knowledge audit."
2	2	"Is able to align business strategy and knowledge management strategy."
3	3	"Is able to use and evaluate knowledge management technologies."
4	4	"Understands the nature of knowledge and socio-ethical and administrative factors of knowledge management."
5	5	"Is able to design knowledge management system."
6	6	"Is able to evaluate the quality of knowledge management artefacts."